

AGRITECH & DATA INSIGHTS PLATFORM

Project Overview

Agriculture-support platform providing soil analysis, crop recommendation, crop disease risk information, weather updates, and market-price transparency.

Problem Statement:	Farmers in rural regions lack real-time digital insights for optimal crop selection and direct market price comparisons.
Solution Implemented:	Developed an accessible agritech portal combining Scikit-Learn soil recommendation models with live weather APIs.
My Technical Role:	Agritech Platform Lead

Main Features & Technical Stack

Core Features:	Technologies Used:
- Soil health parameter analyzer providing ML crop recommendations - Weather forecast and disease risk advisory dashboard - Market price comparison tool across regional mandis - High-contrast multilingual farmer-friendly interface	HTML, CSS, JavaScript, Chart.js, Flask/FastAPI, Python, Scikit-Learn, Render
	Architecture Flow:
	Farmer Web UI → FastAPI Service → Weather API → Scikit-Learn Model → Render Host

Technical Challenge & Results

Key Engineering Challenge: Simplified UI with icon-based navigation for users with low digital literacy.
Validated Results: Created a comprehensive agricultural decision-support platform.

Live Demo: <https://github.com/prashant-singh-78/krashi-kalyan.git> | GitHub: <https://github.com/prashant-singh-78/krashi-kalyan.git> | Portfolio: [prashant08.vercel.app](#)